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Extremal Animals

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We regard triangular, square and hexagonal animals as plane graphs with p points, q lines and n cells. It is easy to determine the maximum value of q in terms of n for each of these three species of animal as these are realized by the animals whose skeleton is a tree. We also established the exact minimum value of q with respect to n by a certain spiral construction. As corollaries we find the five other pairs of upper and lower bounds of the three animal parameters p, q, n in terms of each other.

In the plane a configuration of nonoverlapping congruent regular polygons is called an *animal* (see [3] and the references cited therein) if each polygon (or cell) is joined together with at least one other polygon along a side, and it is simply connected, that is, there are no holes in its interior. As there are exactly three space-filling regular polygons, we will distinguish the corresponding triangular, square and hexagonal animals (the three *species* of animals). One of each type is shown in Figure 1.

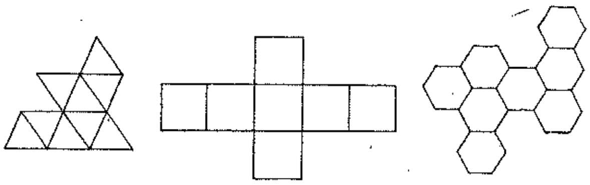

Figure 1.

A triangular, a square, and a hexagonal animal

It is an unsolved enumeration problem to determine the number of different animals with n cells (see for example [2]). Here we will investigate the following more tractable questions concerning extremal properties of different animals.

Every animal can be regarded as a plane graph (see [1]) with p points, q lines, and n cells. Given n, what are maximum and minimum values of

q and of p? Which p and q are possible between these maximum and minimum values? Which animals attain the extremal values of p and q? Let a denote the polygonal length of an animal.

THEOREM 1. *For triangular* $(a = 3)$, *square* $(a = 4)$, *and hexagonal* $(a = 6)$ *animals,*

$$max\ q = (a - 1)n + 1. \tag{1}$$

An animal attains this maximum number of lines if and only if its skeleton is a tree.

Proof. Let q_1 and q_2 denote the number of lines which belong to one and two cells, respectively, so that

$$q = q_1 + q_2. \tag{2}$$

By counting the sides of all the cells in an animal, we get

$$na = q_1 + 2q_2 = q + q_2, \tag{3}$$

from which

$$q = na - q_2. \tag{4}$$

We shall find that equation (4) will be quite useful in determining min q for all three species of animals.

The *skeleton* of an animal is defined as its inner dual. That is the graph obtained when we take the center of each cell as a point, and join a pair of these points by a line if and only if their cells have a side in common. Thus the skeleton of an animal is a connected graph with n points and q_2 lines. Clearly the skeleton of an animal has the minimum number $q_2 = n - 1$ of lines if and only if it is a tree. By (4) and by the fact that animals with tree-skeletons (for example with path P_n) do exist, the theorem is proved. ∎

Several open problems suggest themselves. Enumerate all animals with max q. Two animals are regarded as equal if they are congruent by translation, rotation or reflection. How many animals have a given tree as skeleton? For paths P_n this question was studied in [6] and [4]. Which

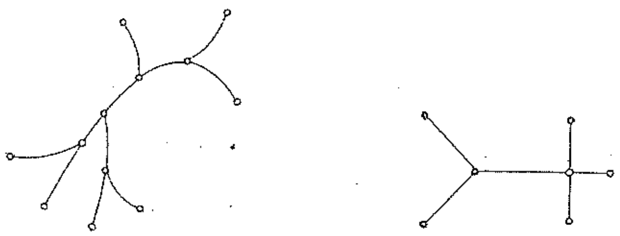

Figure 2.
Trees not being the skeleton of triangular or hexagonal, and of square animals.

trees can be the skeleton of some animal? The condition for triangular or hexagonal animals that the maximum degree not exceed three (four for square animals) is necessary but not sufficient (see Figure 2).

In the following $\{x\}$ shall denote the smallest integer $\geqslant x$ (the "waiter symbol"). In the next three theorems we determine the minimum number of lines in square, hexagonal and triangular animals respectively.

THEOREM 2. *For square animals with n cells, the minimum number of lines is:*

$$\min q = 2n + \{2\sqrt{n}\}. \tag{5}$$

Proof. By Euler's polyhedron formula, the number f of squares contained in the skeleton of an animal is:

$$f = q_2 - n + 1. \tag{6}$$

Let q_{2i} denote the number of lines of the skeleton which are sides of i squares ($i = 0, 1, 2$). Counting all sides of the squares we obtain

$$4f = q_{21} + 2q_{22} = q_2 - q_{20} + q_{22}. \tag{7}$$

From (6) and (7), we get

$$q_2 = q_2(n) = \tfrac{4}{3}(n-1) - \tfrac{1}{3}q_{20} + \tfrac{1}{3}q_{22}. \tag{8}$$

In addition we define $q_2(0) = 0$. Then for small values of n,

$$q_2(n) \leqslant 2n - \{2\sqrt{n}\} \tag{9}$$

is easily seen to be valid. To prove (9) by induction on n, we consider the f squares of the skeleton as cells and observe that $q_{22} \leqslant \max q_2(f)$. From (8) and (9) it follows that

$$q_2 \leqslant \tfrac{4}{3}(n-1) + \tfrac{1}{3}q_{22} \leqslant \tfrac{4}{3}(n-1) + \tfrac{1}{3}(2f - \{2\sqrt{f}\}). \tag{10}$$

Together with (6) we get

$$q_2 \leqslant 2n - 2 - 2\sqrt{q_2 - n + 1}. \tag{11}$$

As equality in (11) holds for $q_2 = 2n \pm 2\sqrt{n}$, and as $q_2 < 2n$ follows from (3), we have proved (9), and thus by (4),

$$\min q \geqslant 2n + \{2\sqrt{n}\}. \tag{12}$$

We now consider those square animals which grow like a spiral

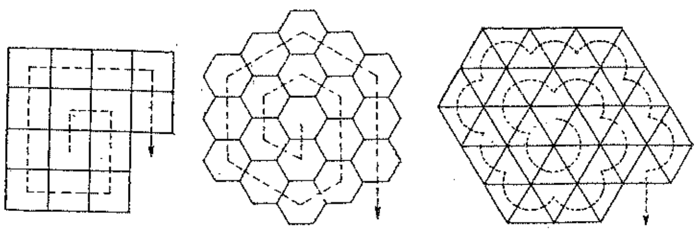

Figure 3.
Animals with minimum number of lines

(Figure 3). The number of lines in such a square animal is given by two formulas depending on the following expressions of n in terms of $k \geqslant 0$:

$$q(n) = \begin{cases} 2k(k+1)+3+2(i-1) & \text{if } n = k^2+i,\ 1 \leqslant i \leqslant k, \\ 2k(k+1)+2k+4+2(i-1) & \text{if } n = k^2+k+i,\ 1 \leqslant i \leqslant k+1. \end{cases} \tag{13}$$

By checking that these values always may be written in the form $2n + \{2\sqrt{n}\}$ we can conclude that

$$\min q \leqslant 2n + \{2\sqrt{n}\}. \tag{14}$$

Finally (12) and (14) together yield (5). ∎

In the case of hexagonal animals, the following result of [5] is useful.

LEMMA 3. *For any packing of n congruent circles in the plane, the maximum number of points where two circles touch each other is:*

$$B(n) = 3n - \{\sqrt{12n-3}\}. \tag{15}$$

THEOREM 3. *For hexagonal animals with n cells, the minimum number of lines is:*

$$\min q = 3n + \{\sqrt{12n-3}\}. \tag{16}$$

Proof. We consider the inscribed circles of all the hexagons of a hexagonal animal with a minimum number of lines. These n circles form a packing. Every tangent point of two circles corresponds uniquely to one of those q_2 lines which are sides of two hexagons. Thus max $q_2 = B(n)$, and with (4) we get min $q = 6n - B(n)$, which by (15) yields (16). ∎

THEOREM 4. *For triangular animals with n cells, the minimum number of lines is:*

$$\min q = n + \{\tfrac{1}{2}(n + \sqrt{6n})\}. \tag{17}$$

Proof. We note that every face of the skeleton of a triangular animal is a hexagon. Following the proof of Theorem 2 by suitably changing the meaning of the notation, we deduce in place of (8) that

$$q_2 = q_2(n) = \tfrac{6}{5}(n-1) - \tfrac{1}{5}q_{20} + \tfrac{1}{5}q_{22}. \tag{18}$$

Now $q_{22} = q_{22}(n)$ is not diminished if it is replaced by max $q_2(f)$ for hexagonal animals with f cells (again defining $q_2(0) = 0$). Applying (4) and (16), it follows that

$$q_{22} \leqslant 3f - \{\sqrt{12f-3}\}. \tag{19}$$

Inserting (19) and (6) into (18), we obtain

$$2q_2 \leqslant 3n - 3 - \sqrt{12q_2 - 12n + 9}. \tag{20}$$

Determining the two values of q_2 when equality holds in (20), and observing that $q_2 < \frac{3}{2}n$ follows from (3) with $a = 3$, we conclude that

$$q_2 \leqslant 2n - \{\tfrac{1}{2}(n + \sqrt{6n})\}. \tag{21}$$

Thus by (4) we have shown that

$$\min q \geqslant n + \{\tfrac{1}{2}(n + \sqrt{6n})\}. \tag{22}$$

Now we consider triangular animals which grow like a spiral (Figure 3). The number of lines in such a triangular animal is now given by three formulas which depend on the following expressions of n in terms of $k \geqslant 0$:

$$q(n) = \begin{cases} 9k^2 + 3k, \text{ if } n = 6k^2, \\ 9k^2 + 3k + i + 1 + [\frac{i+1}{2}], \text{ if } n = 6k^2 + i, 1 \leqslant i \leqslant 2k \\ 9k^2 + 3k + j(2k+1) + i + k + (j-1)(k+1) + [\frac{i+1}{2}], \\ \quad \text{if } n = 6k^2 + j(2k+1) + i - 1, 1 \leqslant i \leqslant 2k+1, 1 \leqslant j \leqslant 5. \end{cases} \tag{23}$$

From this we get

$$\min q \leqslant n + \{\tfrac{1}{2}(n + \sqrt{6n})\}, \tag{24}$$

as the values of the three different formulas in (23) can be checked to be equal to the right hand side of (24). Then (22) together with (24) complete the proof. ∎

Now that the minimum and maximum number of lines have been determined precisely for all three species of animals with n cells, we next derive an interpolation theorem which proves that animals always exist for all values of q between these two extremes. For the three parameters n, p, q associated with each animal, q has been bounded in terms of n in Theorems 1–4. Corresponding bounds for q in terms of p, for p in terms of n and of q, and for n in terms of p and of q are given in Corollaries 1–3 below.

THEOREM 5. *For any $q = q(n)$ between min q and max q there exists at least one animal with n cells and q lines.*

Proof. The assertion obviously holds for small values of n. We assume the validity for n and add one further cell with only one common side. Then by addition of $a - 1$ to (1), (5), (16), (17), we obtain animals for all values $q = q(n + 1)$, except perhaps for min $q(n + 1)$, and $1 + \min q(n + 1)$ in case $a = 6$. However, animals with min q lines do exist (Figure 3), and for $n \geqslant 2$ we always can add a hexagon with exactly two common sides to the hexagonal animal of spiral type with n cells. ∎

Which values of p are possible for given n? If the number p ($\geqslant a$) of points or q ($\geqslant a$) of lines are given, for which n and q, or n and p, do there exist animals? These questions may be answered immediately by using

Euler's formula,

$$p + n = q + 1, \tag{25}$$

as in Theorems 1, 2, 3, 4 and 5. The results are listed in the following corollaries.

COROLLARY 1. *Triangular animals exist if and only if p, q and n are within the ranges*

$$1 + \{\tfrac{1}{2}(n + \sqrt{6n})\} \leqslant p \leqslant n + 2 \tag{26}$$

$$1 + \{\tfrac{1}{3}(q - 1 + \sqrt{4q+1})\} \leqslant p \leqslant q + 1 - \{\tfrac{1}{2}(q-1)\} \tag{27}$$

$$2p - 3 \leqslant q \leqslant 3p - \{\sqrt{12p-3}\} \tag{28}$$

$$p - 2 \leqslant n \leqslant 2p + 1 - \{\sqrt{12p-3}\} \tag{29}$$

$$\{\tfrac{1}{2}(q-1)\} \leqslant n \leqslant q - \{\tfrac{1}{3}(q - 1 + \sqrt{4q+1})\} \tag{30}$$

COROLLARY 2. *Square animals exist if and only if p, q and n are within the ranges*

$$n + 1 + \{2\sqrt{n}\} \leqslant p \leqslant 2n + 2, \tag{31}$$

$$1 + \{\tfrac{1}{2}(q - 1 + \sqrt{2q+1})\} \leqslant p \leqslant q + 1 - \{\tfrac{1}{3}(q-1)\}, \tag{32}$$

$$p - 1 + \{\tfrac{1}{2}(p-2)\} \leqslant q \leqslant 2p - \{2\sqrt{p}\}, \tag{33}$$

$$\{\tfrac{1}{2}(p-2)\} \leqslant n \leqslant p + 1 - \{2\sqrt{p}\}, \tag{34}$$

$$\{\tfrac{1}{3}(q-1)\} \leqslant n \leqslant q - \{\tfrac{1}{2}(q - 1 + \sqrt{2q+1})\}. \tag{35}$$

COROLLARY 3. *Hexagonal animals exist if and only if p, q and n are within the ranges*

$$2n + 1 + \{\sqrt{12n-3}\} \leqslant p \leqslant 4n + 2, \tag{36}$$

$$1 + \{\tfrac{1}{3}(2q - 2 + \sqrt{4q+1})\} \leqslant p \leqslant q + 1 - \{\tfrac{1}{5}(q-1)\}, \tag{37}$$

$$p - 1 + \{\tfrac{1}{4}(p-2)\} \leqslant q \leqslant 2p - \{\tfrac{1}{2}(p + \sqrt{6p})\}, \tag{38}$$

$$\{\tfrac{1}{4}(p-2)\} \leqslant n \leqslant p + 1 - \{\tfrac{1}{2}(p + \sqrt{6p})\}, \tag{39}$$

$$\{\tfrac{1}{5}(q-1)\} \leqslant n \leqslant q - \{\tfrac{1}{3}(2q - 2 + \sqrt{4q+1})\}. \tag{40}$$

We now find some values of p and q for which there do not exist animals of each species.

THEOREM 6. *For $p, q > a$, there do not exist animals just when*

(1) $q = 4, 6, 8, 10$ *for triangular animals*

(2) $p = 5, 7$ *or* $q = 5, 6, 8, 9, 11, 14$ *for square animals*

(3) $p = 7, 8, 9, 11, 12, 15$ *or* $q = 7, 8, 9, 10, 12, 13, 14, 17, 18, 22$ *for hexagonal animals.*

Proof. For every case we determine n_0 such that $\min p(n) \leqslant 1 + \max p(n-1)$ when $n \geqslant n_0$ and the corresponding inequality for q. Then (26), (1) and (17), (31), (1) and (5), (36), (1) and (16) together with the remaining $n < n_0$ permit us to find the desired values. ∎

THEOREM 7. *The collected results still remain true if in addition holey animals are allowed.*

Proof. If a hole has the size of one cell, this may be filled up by this one cell and the number of lines and points is unchanged. Otherwise we find on the boundary of triangular animals at least two neighboring triangles which can be put into a hole such that the number of lines or points does not increase. On the boundary of a square or hexagonal animal we can find at least one cell with at least two or three lines or at least one or two points which belong only to this cell. Within any hole, there is at least one place where a new cell needs at most two or three new lines, or at most one or two new points. Thus by filling up the holes, no holey animal has fewer than min q lines or min p points.

In the case of maximum we open the holes by deleting partial animals having a path as skeleton. If we then connect these paths only by one end line to the boundary of the original animal, the number of lines or points is always increased. ∎

We also could classify the animals (without holes) by their different circumferences q_1. From (2) and (3) we see that

$$q_1 = 2q - an. \tag{41}$$

Thus (1), (5), (16), (17) and Theorem 5 enable us at once to get extremal as well as interpolated values of q_1.

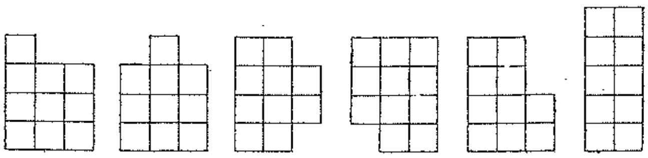

Figure 4.
Square Animals with 10 cells and min $q = 27$ lines

Examples of animals with minimum circumference q_1, min q (lines), or min p (points) are shown in Figure 3. It is not difficult to prove that these, in the case of square animals for instance, are unique if the number of cells is a perfect square ($n = k^2$). However, there are other values of n with more than one animal (see Figure 4). How many different animals with n cells and min q lines are there? This may be a tractable problem on animal enumeration unlike those proposed in [2].

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